

Q13

Diffraction rating formula: $n\lambda = d\sin\theta_1$

Consider $n = 1$. Then $5.30 \times 10^{-7} = d\sin 15.4^\circ$ or $d = 1.996 \times 10^{-6} \text{ m}$

For $n = 2$, we have $2\lambda = d\sin\theta_2$

$$2 \times 5.30 \times 10^{-7} = 1.996 \times 10^{-6} \sin\theta_2$$

$$\theta_2 = 32.1^\circ.$$

Hence, $\theta_2 - \theta_1 = 16.7^\circ$

Ans: C

Q14